Evaluating LLMs for Automated SEM Powder Characterization Workflows

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Abstract

Powder morphology is critical to powder-based additive manufacturing, where particle size and shape influence deposition behavior and final part quality. While computer vision methods can automate SEM particle segmentation and morphology extraction, tailoring downstream analyses to specific characterization goals still frequently requires users to select or modify specialized analysis tools. To address this limitation, we evaluate LLM-guided automated characterization, where large language models (LLMs) interpret natural-language characterization requests using three strategies: raw-data inference, precomputed-summary retrieval, and deterministic-tool routing. Across 13 API-accessible and locally hosted LLM configurations and 93,379 characterization tasks, models achieved 92.0% accuracy when retrieving measurements from precomputed summaries, compared with 48.8% when computing them from raw particle records. For requests requiring additional analysis, schema-guided routing allowed models to select deterministic tools and arguments rather than compute directly, with 10 of 13 configurations scoring at least 95.0% routing accuracy. However, performance declined as tasks became more open-ended, indicating that LLM-guided characterization is most reliable for bounded analysis requests while open-ended morphology selection remains challenging.

1 Introduction

Motivation. Powder morphology is critical to powder-based additive manufacturing, where changes in particle size, shape, and distribution can influence flowability, deposition behavior, and final part performance [7, 18, 19]. Scanning electron microscopy (SEM) is commonly used to characterize these properties because it provides high-resolution images of individual particles and their surface morphology. However, manual extraction of particle-level measurements from dense SEM images is time-consuming, difficult to scale, and sensitive to annotator variability.

Computer Vision for Automated Powder Characterization. Recently, researchers have applied computer vision (CV) methods to reduce the manual effort required for SEM-based powder analysis. For example, Cohn et al. used instance segmentation to identify individual particles and satellites [2], while Price et al. explored active learning to improve performance across powder types and imaging magnifications [12]. More recently, DualSight combined a lightweight segmentation model with promptable foundation-model refinement to improve particle boundaries [10], allowing high-quality particle segmentations to be produced at scale.

Limitations of Automated Characterization Frameworks. While CV methods can automate particle labeling and morphology extraction, configuring these outputs for a specific materials analysis still frequently requires manual intervention. A single SEM image may contain hundreds of segmented particles from which size and morphology descriptors are extracted and aggregated into image- or batch-level statistics. Further, the measurements needed depend on the characterization

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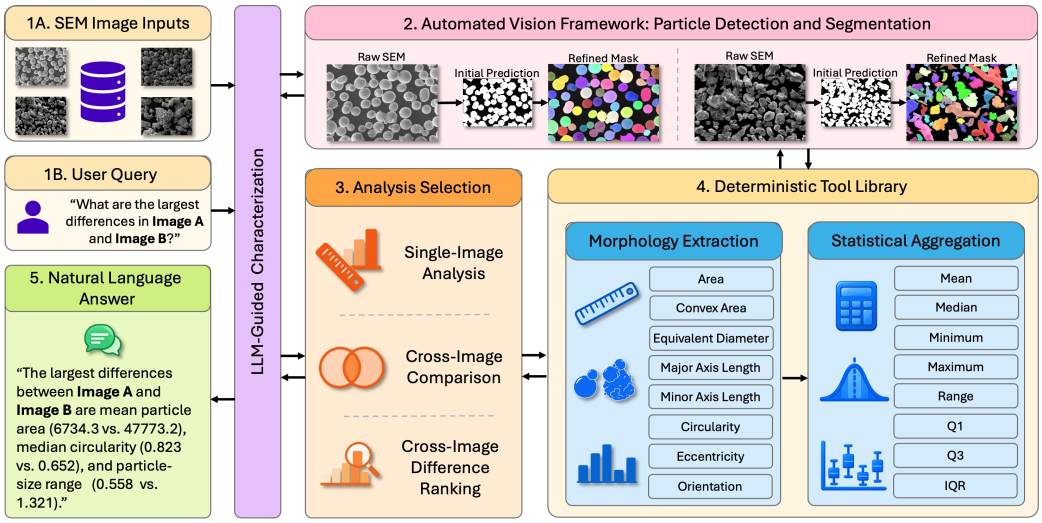


Figure 1: **LLM-guided characterization workflow with tool routing.** An automated vision pipeline segments SEM images, while the LLM interprets the user request and selects image(s), analysis type, morphology feature, and statistic. Deterministic morphology-extraction and statistical-aggregation tools compute the result, which is returned as a natural-language response.

objective. For example, one analysis may prioritize particle-size statistics such as D10, D50, and D90 from a single image, while another may focus on cross-image differences in shape characteristics. As a result, practitioners must often manually select or modify study-specific analysis code as characterization objectives change.

Our Work. To address this challenge, we evaluate LLM-guided automated characterization, where LLMs interpret natural-language requests and select available morphology measurements or deterministic analysis tools (Fig. 1). Building on existing CV-based powder characterization and recent scientific tool-use work [5, 6], we compare three strategies for providing LLMs access to SEM-derived morphology data and analysis tools: (1) direct inference from raw particle data, (2) retrieval from precomputed morphology summaries, and (3) schema-guided analysis routing to deterministic tools [1, 9, 13]. We further evaluate these strategies across increasingly open-ended materials-characterization tasks, ranging from single-image statistics to cross-image comparisons and morphology-difference identification. Our contributions include:

- We evaluate 13 API-accessible and locally hosted LLM configurations across 93,379 tasks, where retrieval from precomputed morphology summaries achieves 92.0% accuracy compared with 48.8% for direct LLM computation from raw particle records.
- We test automated analysis routing to deterministic morphology tools, with 10 of 13 model configurations reaching at least 95.0% routing accuracy.
- We evaluate increasingly open-ended characterization tasks, where pairwise accuracy improves from 72.5% on raw records to 94.3% with precomputed values, but exact-metric top-3 selection only scores 5.6%, showing that morphology selection remains challenging.

2 Methodology

Characterization Workflow and Data. For this evaluation, we use a publicly accessible powder SEM dataset containing 7,324 segmented particles across 132 SEM images and 13 powder types [10]. We use a DualSight-style two-stage segmentation pipeline, where YOLOv8-Nano [16] creates an initial segmentation and Segment Anything Model (SAM) [3] is used for segmentation refinement [4, 10, 11]. Refined masks are converted into per-particle polygons, from which Scikit-Image [17] extracts particle-level size and shape descriptors. For the primary benchmarks, we use a 20-image held-out evaluation set spanning all 13 powder types and score LLM responses against deterministic

reference values computed from extracted particle measurements. *We release the evaluation split, routing schemas, and evaluation code at* https://github.com/sprice134/llms_for_cv.

Benchmark Design. We evaluate each LLM configuration across five studies covering direct inference, materials-specific characterization, analysis routing, repeatability, and increasingly open-ended characterization. Our primary direct-inference evaluation contains 5,120 tasks, spanning eight particle descriptors, eight summary statistics, four input representations, and 20 held-out SEM images. Inputs use either raw per-particle measurements or precomputed morphology summaries in text or tabular form. Raw inputs require the LLM to compute the requested statistic directly [20], while processed inputs provide precomputed deterministic summary values, testing if the model can retrieve the requested measurement without performing the numerical calculation. In total, we evaluate 7,183 tasks per model configuration and 93,379 evaluations across all 13 configurations. Complete task definitions are provided in Appendix A, with prompt templates in Appendix C.

LLM Model Selection and Evaluation. We evaluate six API-accessible GPT configurations and seven locally hosted open-weight configurations spanning model scale, dense and mixture-of-experts architectures, and reasoning settings. Complete model configurations and inference details are provided in Appendix A.2. Numeric responses are treated as correct when within 1% of the deterministic reference value to account for rounding differences. For schema-guided analysis routing, we evaluate both exact call matching and functionally equivalent outputs, similar to existing function-calling evaluations [8]. Strict scoring requires the exact reference tool and arguments, while accepted scoring allows predefined alternatives known to produce the same deterministic result. For example, D_{50} may be accepted in place of median diameter. We report model-macro averages with percentile 95% confidence intervals from 20,000 model-family cluster-bootstrap resamples.

3 Experimental Results and Analysis

Direct Answering & Input Representation. As shown in Table 1 and Fig. 2, across models, LLMs achieved an average direct-answer accuracy of 92.0% when using precomputed morphology summaries, with several configurations scoring 100.0% on processed tables and 99.5% on processed text. In comparison, accuracy on raw particle records averaged 48.8%, resulting in a 43.2 percentage-point paired difference (95% CI: 36.1–52.6 pp). The materials-specific characterization study showed a similar pattern, with precomputed values reaching 100.0% accuracy on single-image numeric questions and 78.4% on pairwise comparisons, compared with 46.8% and 65.6% from raw inputs. Relaxing the 1% tolerance to 5% increased raw accuracy from 48.8% to 58.7%, indicating that the remaining errors are not simply due to rounding or numerical precision. Performance also varied

Table 1: **LLM performance across SEM characterization tasks.** Open-weight and GPT-model results (%) across direct inference, analysis routing, repeatability, and comparative characterization. Values report task accuracy, response agreement, or top-3 precision, as defined by column. Additional information is provided in Appendix A. Higher values are better, with the best value per metric **bolded** and second-best underlined.

	Direct answer				Tool call		Repeatability		Case study	
Model	Raw table	Raw text	Proc. table	Proc. text	Accepted call	Strict call	Repeat accuracy	Repeat agreement	Pairwise comparison	Family top-3
Open-weight / local										
Gemma4 31B	7.0	13.4	68.3	98.5	**100.0**	94.4	46.0	93.4	73.9	32.2
Qwen3.6-35B-A3B	78.6	72.0	**100.0**	**99.5**	99.8	92.4	91.5	99.6	90.6	38.9
GPT-OSS-20B (Low)	53.4	52.2	98.4	**99.5**	97.1	90.1	74.7	82.0	86.7	35.6
GPT-OSS-20B (Medium)	54.7	54.3	98.4	**99.5**	96.4	88.3	74.7	82.8	87.8	36.7
GPT-OSS-20B (Default)	54.9	53.5	97.6	**99.5**	95.5	87.7	74.2	84.0	86.7	28.9
Gemma4-26B-A4B	12.7	9.2	62.0	50.1	95.5	76.8	28.6	88.3	60.6	3.3
Bonsai-8B	5.9	9.8	65.5	82.7	36.4	27.0	41.8	**100.0**	65.0	30.0
GPT series / API										
GPT-5.6-Luna (Medium)	**90.1**	**88.0**	**100.0**	**99.5**	**100.0**	93.3	**94.4**	74.2	93.3	45.6
GPT-5.6-Luna (Instant)	43.4	43.3	**100.0**	**99.5**	**100.0**	**94.6**	73.4	74.2	90.6	**51.1**
GPT-5.4-Mini (Medium)	89.8	85.8	**100.0**	**99.5**	**100.0**	85.6	93.9	77.0	**94.4**	38.9
GPT-5.4-Mini (Instant)	42.2	45.8	**100.0**	**99.5**	**100.0**	92.1	73.7	79.7	82.2	41.1
GPT-5.4-Nano (Medium)	73.0	79.1	95.6	97.1	85.4	74.4	87.2	74.2	91.7	36.7
GPT-5.4-Nano (Instant)	25.0	31.3	89.8	91.1	60.0	52.4	59.5	80.5	80.6	47.8

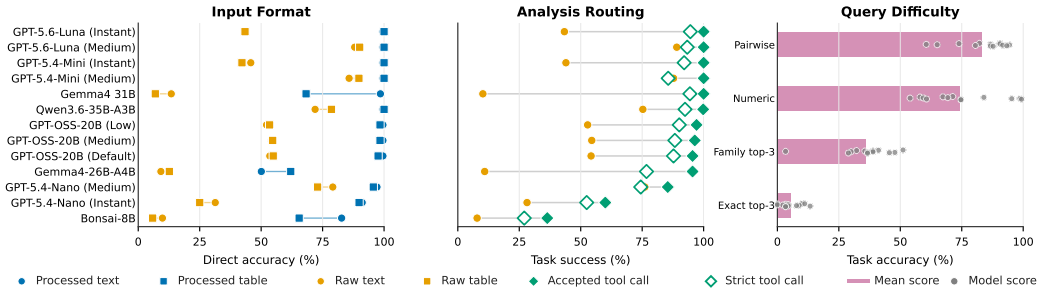


Figure 2: **LLM performance across characterization tasks.** *Left:* direct accuracy across raw and processed text and tabular input serializations. *Center:* accepted and strict analysis-routing accuracy. *Right:* performance across pairwise, single-image numeric, and top-3 morphology comparisons. Gray points show individual model scores and bars show model-macro averages.

by statistic, with raw minimum and maximum queries averaging 78.1% and 79.2%, compared with 17.6% for IQR. At the model level, `GPT-5.6 Luna (Medium)` achieved the strongest raw-input performance at 90.1% on tables and 88.0% on text, while `Qwen3.6-35B-A3B` remained the strongest open-weight configuration at 75.3% averaged across raw representations.

Schema-Guided Analysis Routing. Under strict scoring, which requires the exact reference tool and arguments, 80.7% of outputs were correct across models. When the same outputs were evaluated under a more relaxed criterion, where alternative arguments producing the same deterministic result are also treated as acceptable, the proportion scored as acceptable increased to 89.7%, a 9.0 percentage-point gain. Using accepted scoring, 5 configurations reached 100.0% accuracy and 10 of 13 scored at least 95.0%. Separated by task type, accepted routing averaged 95.9% for single-image requests and 96.2% for top-k analysis, compared with 83.1% for pairwise requests. Pairwise routing errors frequently occurred in weaker model configurations and most commonly stemmed from malformed arguments, incorrect metric selection, or incorrect statistic selection.

Repeatability & Comparative Characterization. High output reproducibility did not guarantee high accuracy. For example, `Qwen3.6-35B-A3B` achieved 99.6% exact-response agreement with 91.5% accuracy, while `Bonsai-8B` produced 100.0% agreement but only 41.8% accuracy. In the comparative case study, pairwise accuracy increased from 72.5% on raw records to 94.3% with precomputed values. In comparison, top-3 precision reached 35.9% at the morphology-family level and 5.6% for exact metric selection, showing that models had difficulty identifying the largest non-redundant morphology differences across 108 candidate metrics.

4 Conclusions & Future Work

We evaluate LLM-guided characterization for automated SEM powder workflows. Across tasks, LLMs were most reliable when characterization results were available through precomputed summaries or schema-guided routing to deterministic analysis tools, while direct computation from dense particle records was less consistent across models. Analysis routing performed well across most model configurations, with 10 of 13 reaching at least 95.0% routing accuracy, showing that LLMs can reliably map a range of bounded characterization needs to deterministic tools. However, performance declined across increasingly open-ended tasks, highlighting that broader morphology selection remains challenging.

These results support a hybrid workflow where computer vision and deterministic analysis perform particle segmentation, morphology extraction, and numerical computation, while LLMs interpret natural-language characterization needs and map them to available measurements or analysis tools. However, our evaluation uses a single SEM dataset and characterization pipeline, so performance on independently collected images or alternative analysis workflows remains untested. Future work should extend these evaluations to additional materials and imaging conditions, compare schema-guided routing with more general-purpose LLM-generated code execution, and further evaluate open-ended characterization needs to better define where LLM-guided characterization can reliably support broader automated workflows, such as autonomous labs [14, 15].

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A Benchmark Design and Evaluation Details

A.1 Evaluation Tasks

In this work, we evaluate each model configuration across five complementary studies spanning direct inference, materials-specific characterization, analysis routing, repeatability, and increasingly open-ended characterization, as summarized in Table 2.

Table 2: **Evaluation task inventory.** Number and scope of tasks evaluated for each model configuration. The five suites contain 7,183 tasks per configuration and 93,379 evaluations across all 13 configurations.

Suite	Tasks	Task definition
Direct inference	5,120	8 descriptors $\times$ 8 statistics $\times$ 4 representations $\times$ 20 images
Materials-specific characterization	380	Single-image statistics and pairwise powder-characterization
Analysis routing	555	225 single-image, 270 pairwise, and 60 top-k tasks
Repeatability	768	256 stratified prompts evaluated three times each
Comparative characterization	360	150 single-image, 180 pairwise, and 30 top-3 comparisons
Total	**7,183**	Per model configuration

A.1.1 Direct Model Inference

In our primary direct-inference evaluation, we evaluate each model configuration across 5,120 tasks spanning 20 SEM images, eight particle characteristics, eight statistical measurements, and four input representations. We use a 2×2 factorial design across text versus tabular inputs and raw particle records versus precomputed morphology summaries, allowing us to compare direct numerical inference with summary retrieval for the same model, image, and requested statistic. The eight particle characteristics include `area`, `area_convex`, `major_axis_length`, `minor_axis_length`, `eccentricity`, `orientation`, `equivalent_diameter`, and `circularity`, spanning various notations and magnitudes. For each characteristic, we evaluate mean, median, range, minimum, maximum, Q1, Q3, and interquartile range (IQR), spanning different computational requirements from simple extrema and averages to distributional statistics requiring ordered particle measurements.

A.1.2 Materials-Specific Characterization

To extend the direct-inference evaluation toward common powder-characterization tasks, we evaluate an additional 380 tasks using materials-specific metrics and terminology. Single-image questions include D_{10}, D_{50}, and D_{90} equivalent diameter, median circularity, and normalized size-distribution span, $(D_{90} - D_{10})/D_{50}$. We additionally evaluate pairwise questions asking which image has the larger median particle size, wider size distribution, or more circular particle morphology. This study evaluates whether the trends observed in direct inference extend to more application-specific characterization tasks and terminology.

A.1.3 Analysis Routing

To evaluate LLM-guided analysis routing to deterministic characterization tools, we evaluate an additional 555 tasks. This includes 225 single-image tasks, 270 pairwise comparison tasks, and 60 top-k morphology-difference tasks. In this evaluation, models select among deterministic operations to *get* a morphology statistic for a single image, *compare* a specified statistic between two images, or *rank* morphology differences between an image pair. For analysis-routing tasks, prompts provide the available tool schema, image and metric identifiers, and the natural-language characterization objective. Models return a structured JSON call specifying the selected tool and arguments, with the same output evaluated under *strict* and *accepted* scoring criteria. Strict scoring requires the exact reference tool and arguments, while accepted scoring additionally allows predefined alternative arguments that produce the same deterministic result, such as using D_{50} instead of median diameter.

A.1.4 Repeatability

To evaluate model reproducibility, we select a stratified random sample of 256 tasks across experimental study, input representation, and answer type. This includes 236 direct-inference numeric tasks, balanced across processed table, processed text, raw table, and raw text representations with 59 tasks each, in addition to 10 materials-specific numeric and 10 pairwise characterization tasks. Using a temperature of `0.0`, we evaluate each task three times, producing a total of 768 inferences per model configuration. We evaluate both response agreement and accuracy, separating model reproducibility from correctness.

A.1.5 Comparative Characterization

To evaluate increasingly open-ended characterization, we construct an additional 360-task study containing 150 single-image numeric questions, 180 directed pairwise comparisons, and 30 structured top-3 morphology comparisons. Single-image and pairwise tasks evaluate targeted characterization analyses using either raw particle records or precomputed morphology values. For the top-3 task, models are provided deterministic summary values spanning 18 morphology characteristics and six statistical measurements, yielding 108 candidate metrics. Models are asked to identify three non-redundant morphology differences between an image pair and return the selected metrics and corresponding direction of change. This study evaluates how model performance changes as the characterization objective becomes less constrained and the model must select from a larger set of candidate metrics.

A.2 Model Selection and Evaluation Details

A.2.1 Model Configurations

We evaluate eight LLMs across 13 configurations, spanning API-accessible and locally hosted models, dense and mixture-of-experts (MoE) architectures, and varying reasoning settings, as summarized in Table 3. For `GPT-OSS-20B`, we evaluate `Low`, `Medium`, and `Default` reasoning configurations, while the GPT API models are evaluated using both `Instant` and `Medium` reasoning. Within each task family, all models use identical prompts and output-format requirements. Open-weight models are hosted locally on a single NVIDIA RTX Pro 6000B GPU using vLLM with unquantized Hugging Face model weights, while API-accessible GPT models are evaluated using the official OpenAI API, with inference performed in July 2026.

Table 3: **LLM configurations evaluated in this study.** Default indicates that no explicit reasoning setting was applied and the model was evaluated using its default inference behavior.

Model	Access	Architecture	Reasoning setting
GPT-5.6 Luna	API	Undisclosed	Instant
GPT-5.6 Luna	API	Undisclosed	Medium
GPT-5.4 Mini	API	Undisclosed	Instant
GPT-5.4 Mini	API	Undisclosed	Medium
GPT-5.4 Nano	API	Undisclosed	Instant
GPT-5.4 Nano	API	Undisclosed	Medium
Gemma4 31B	Local	Dense	Default
Gemma4-26B-A4B	Local	MoE	Default
Qwen3.6-35B-A3B	Local	MoE	Default
GPT-OSS-20B	Local	MoE	Low
GPT-OSS-20B	Local	MoE	Medium
GPT-OSS-20B	Local	MoE	Default
Bonsai-8B	Local	Dense	Not Applicable

A.2.2 Evaluating Model Performance

Unless otherwise stated, we treat a numeric response as correct when the parsed output is within 1% of the deterministic reference value, allowing for differences in precision and rounding. We further evaluate the sensitivity of this threshold in Sec. B. For analysis routing, the same output is evaluated under two criteria: *strict* scoring requires the exact reference tool and arguments, while *accepted*

scoring additionally allows predefined alternative arguments that produce the same deterministic result, such as median versus the 50th percentile.

For repeatability, we report repeat accuracy as the average accuracy across three responses for each sampled task. To evaluate model agreement, we report exact-response agreement and parsed-answer agreement. Exact-response agreement requires all three whitespace-normalized responses to match, while parsed-answer agreement requires the extracted answers to match across repetitions.

For comparative characterization, single-image and pairwise tasks are scored against deterministic reference answers. For top-3 comparisons, candidate metrics are ranked by their absolute normalized pairwise difference between images, and the reference top-3 is selected across distinct morphology families to avoid redundant selections from closely related descriptors. For example, `equivalent_diameter` and `feret_diameter_max` are grouped as particle-diameter measures. Thus, several highly ranked size metrics cannot occupy multiple reference positions when they describe the same underlying morphology characteristic. Exact P@3 requires matching the specific reference metric identifiers, while family P@3 gives credit when a selected metric belongs to the same predefined morphology family as the reference. Because the reference rankings are constructed from normalized morphology differences rather than independent expert review, both scores measure agreement with the deterministic reference rather than whether alternative metric selections are scientifically informative.

For uncertainty estimates, we compute each metric at the model-configuration level and use a model-family cluster bootstrap with 20,000 resamples. In each resample, the eight model families are sampled with replacement, while all configurations within a selected family are retained together. The eight families include Luna, Mini, Nano, Gemma4 31B, Gemma4-26B-A4B, GPT-OSS-20B, Qwen3.6-35B-A3B, and Bonsai-8B, with reasoning configurations grouped within their corresponding model family. We report the 2.5th and 97.5th percentiles of the resulting distribution as the 95% confidence interval.

B Extended Results

B.1 Direct-Inference Robustness

B.1.1 Sensitivity to Numeric Tolerance

In Table 4, we report direct-inference accuracy averaged across all 13 model configurations under varying numeric tolerance thresholds. Here, relaxing the tolerance from 0.1% to 5% increased raw-text accuracy from 41.9% to 59.1% and raw-tabular accuracy from 43.4% to 58.2%. In comparison, processed inputs remained high, ranging from 88.3-94.3% for text and 90.3-90.8% for tabular representations. Even under the 5% tolerance, raw-input performance scored lower than processed inputs, showing that the remaining errors are not explained by minor differences in rounding or numerical precision.

Table 4: **Direct-inference sensitivity to numeric tolerance.** Accuracy across raw and processed text and tabular representations under increasingly relaxed numeric-error thresholds. Best performance at each threshold is **bolded**.

Direct Input	0.1%	0.5%	1%	5%
Raw text	41.9%	46.3%	49.1%	59.1%
Raw tabular	43.4%	45.8%	48.5%	58.2%
Processed text	88.3%	**93.2%**	**93.5%**	**94.3%**
Processed tabular	**90.3%**	90.3%	90.4%	90.8%

B.1.2 Performance by Requested Summary Statistic

In Table 5, we report direct-inference accuracy across each evaluated statistic using a 1% tolerance threshold. Here, raw-input performance varied substantially by statistic, with minimum and maximum reaching 79.7% and 82.3% for raw text, compared with 16.7% for IQR. In comparison, processed inputs remained consistently high across statistics, with processed text providing the highest accuracy for six of eight metrics and processed tabular performing best for median and range. These results

show that raw-input performance decreases for statistics requiring more computational steps, while precomputed summaries remain comparatively stable, highlighting the difference between model-based computation and direct retrieval.

Table 5: **Direct-inference accuracy by requested statistic.** Accuracy at the 1% numeric-error threshold across raw and processed text and tabular representations. The highest value within each representation is **bolded**. The best representation for a given statistic is marked by *.

Statistic	Raw text	Raw tabular	Processed text	Processed tabular
Mean	54.1%	53.4%	**97.5%***	**95.7%**
Median	46.7%	46.0%	93.7%	94.7%*
Range	55.1%	56.6%	93.5%	93.9%*
Minimum	79.7%	**76.4%**	94.5%*	91.3%
Maximum	**82.3%**	76.0%	94.9%*	92.9%
Q1	26.5%	28.1%	91.9%*	85.1%
Q3	31.3%	33.1%	88.6%*	85.1%
IQR	16.7%	18.4%	93.6%*	84.7%

B.2 Analysis Routing Performance by Task Type

In Table 6, we report strict and accepted routing performance across single-image, pairwise, and top-k characterization tasks. Single-image tasks reached 94.0% under strict scoring and 95.9% under accepted scoring, while top-k analysis routing reached 96.2% under both criteria. In comparison, pairwise tasks reached 66.2% under strict scoring and 83.1% under accepted scoring. Pairwise tasks showed the largest difference between strict and accepted scoring, increasing from 66.2% to 83.1%. Of the 593 outputs that failed strict scoring but passed accepted scoring, models frequently selected alternative but equivalent arguments, such as `elongation` instead of `aspect_ratio` or `percentile_50` instead of `median`. Among outputs that also failed accepted scoring, 463 were non-executable due to malformed arguments, while most remaining failures resulted from incorrect tool or statistic selection, frequently in weaker model configurations. In comparison, top-k routing reached 100.0% under both scoring criteria for 12 of 13 configurations, with `Bonsai-8B` as the only exception.

Table 6: **Analysis-routing performance by task type.** Strict scoring requires the exact reference tool and arguments, while accepted scoring allows predefined alternative arguments that produce the same deterministic result.

	Single-image		Pairwise		Top-k	
Model	Strict	Accepted	Strict	Accepted	Strict	Accepted
Open-weight / local						
Gemma4 31B	99.6	100.0	88.9	100.0	100.0	100.0
Qwen3.6-35B-A3B	100.0	100.0	84.4	99.6	100.0	100.0
GPT-OSS-20B (Low)	100.0	100.0	79.6	94.1	100.0	100.0
GPT-OSS-20B (Medium)	99.6	100.0	76.3	92.6	100.0	100.0
GPT-OSS-20B (Default)	100.0	100.0	74.8	90.7	100.0	100.0
Gemma4-26B-A4B	88.4	95.1	61.9	94.8	100.0	100.0
Bonsai-8B	49.3	53.8	3.3	18.9	50.0	50.0
GPT series / API						
GPT-5.6-Luna (Medium)	100.0	100.0	86.3	100.0	100.0	100.0
GPT-5.6-Luna (Instant)	100.0	100.0	88.9	100.0	100.0	100.0
GPT-5.4-Mini (Medium)	99.1	100.0	71.1	100.0	100.0	100.0
GPT-5.4-Mini (Instant)	97.3	100.0	85.9	100.0	100.0	100.0
GPT-5.4-Nano (Medium)	95.6	99.6	51.1	70.4	100.0	100.0
GPT-5.4-Nano (Instant)	93.3	98.7	7.8	18.9	100.0	100.0
Average	**94.0**	**95.9**	**66.2**	**83.1**	**96.2**	**96.2**

B.3 Inference Repeatability and Response Agreement

In Table 7, we report repeat accuracy and response agreement across three evaluations of each sampled task. Across models, repeat accuracy averaged 70.3%, compared with 83.8% exact-response

agreement and 84.8% parsed-answer agreement. However, higher agreement did not necessarily indicate correctness. `Bonsai-8B` produced identical responses across all repeated tasks but reached only 41.8% repeat accuracy, while `Gemma4-26B-A4B` reached 88.3% exact agreement with only 28.6% accuracy. In comparison, `GPT-5.6-Luna (Medium)` achieved the highest repeat accuracy at 94.4%. In addition to run-to-run variation, several open-weight configurations failed to produce a complete response after reaching their maximum generation length, including 24 outputs for `Qwen3.6-35B-A3B` and between 34 and 56 outputs across the `GPT-OSS-20B` configurations. For `GPT-OSS-20B`, incomplete responses increased from 34 under Low reasoning, 43 under default reasoning, and 56 under Medium reasoning, showing that reasoning configuration can also affect completion reliability and should be considered when selecting a model for characterization workflows, particularly for open-weight models. By task type, processed table and text inputs reached 87.6% and 93.7% repeat accuracy, compared with 50.0% and 47.5% for raw inputs, respectively. These results show that high response agreement can reflect either reliable performance or repeated incorrect answers.

Table 7: **Repeatability across model configurations.** Repeat accuracy includes all 768 responses, while parsed-only accuracy is calculated over successfully parsed responses. Exact agreement measures identical normalized responses across three repetitions, while parsed agreement compares the extracted answers.

Model	Completed	Parsed	Repeat Acc.	Parsed Acc.	Exact Agree	Parsed Agree
Open-weight / local						
Gemma4 31B	768/768	768/768	46.0%	46.0%	93.4%	93.4%
Qwen3.6-35B-A3B	744/768	744/768	91.5%	**94.5%**	99.6%	96.5%
GPT-OSS-20B (Low)	734/768	734/768	74.7%	78.2%	82.0%	84.8%
GPT-OSS-20B (Medium)	712/768	711/768	74.7%	80.7%	82.8%	81.2%
GPT-OSS-20B (Default)	725/768	723/768	74.2%	78.8%	84.0%	81.6%
Gemma4-26B-A4B	768/768	768/768	28.6%	28.6%	88.3%	88.7%
Bonsai-8B	768/768	768/768	41.8%	41.8%	**100.0%**	**100.0%**
GPT series / API						
GPT-5.6-Luna (Medium)	768/768	768/768	**94.4%**	94.4%	74.2%	80.5%
GPT-5.6-Luna (Instant)	768/768	768/768	73.4%	73.4%	74.2%	76.6%
GPT-5.4-Mini (Medium)	768/768	768/768	93.9%	93.9%	77.0%	80.1%
GPT-5.4-Mini (Instant)	768/768	768/768	73.7%	73.7%	79.7%	80.9%
GPT-5.4-Nano (Medium)	768/768	768/768	87.2%	87.2%	74.2%	78.1%
GPT-5.4-Nano (Instant)	768/768	768/768	59.5%	59.5%	80.5%	80.5%

B.4 Comparative Characterization Error Analysis

As shown in Table 8, processed inputs improved pairwise accuracy across all three characterization objectives, increasing overall accuracy from 72.5% to 94.3%. The largest improvement was observed for normalized size-distribution comparisons, which increased from 59.2% to 91.3%, a 32.1 percentage-point gain. Comparing the similarity of the paired characterization values across correct and failed comparisons, we found that failed comparisons involved more similar values, with a median difference of 15.9% compared with 25.4% for correct comparisons. Analyzing the 389 failed responses, 378 (97.2%) returned a structurally valid but incorrect response of A, B, or tie, while only 11 resulted from empty, error, or unparsable outputs.

Table 8: **Comparative characterization performance by input condition and objective.** Pairwise accuracy across raw and processed inputs, with median differences for correct and failed comparisons.

	Accuracy			Median difference	
Objective	Raw	Processed	Δ	Correct	Failed
Larger median particle size	76.2%	93.6%	+17.4 pp	46.4%	26.0%
Wider normalized size distribution	59.2%	91.3%	+32.1 pp	24.7%	15.9%
More circular particle morphology	82.1%	97.9%	+15.9 pp	14.7%	9.5%
Overall	**72.5%**	**94.3%**	**+21.8 pp**	**25.4%**	**15.9%**

At the model level, Table 9 shows that processed pairwise inputs reached at least 97.8% accuracy for 11 of 13 configurations, with several reaching 100.0%. In comparison, performance decreased significantly for the open-ended top-3 task. Exact-metric P@3 averaged 5.6% and scored below 14% for every configuration, while morphology-family P@3 reached 35.9%. These results show

that models can reliably compare specified morphology measurements when deterministic values are provided, but selecting the highest-ranked morphology differences from a larger set of candidate metrics remains challenging.

Table 9: **Comparative characterization performance across model configurations.** Pairwise accuracy across raw and processed inputs and performance on open-ended top-3 morphology selection.

	Pairwise accuracy			Open-ended top-3	
Model	Raw	Processed	Δ	Exact P@3	Family P@3
Open-weight / local					
Gemma4 31B	50.0%	97.8%	+47.8 pp	0.0%	32.2%
Qwen3.6-35B-A3B	81.1%	100.0%	+18.9 pp	3.3%	38.9%
GPT-OSS-20B (Low)	73.3%	100.0%	+26.7 pp	4.4%	35.6%
GPT-OSS-20B (Medium)	75.6%	100.0%	+24.4 pp	3.3%	36.7%
GPT-OSS-20B (Default)	75.6%	97.8%	+22.2 pp	3.3%	28.9%
Gemma4-26B-A4B	57.8%	63.3%	+5.6 pp	0.0%	3.3%
Bonsai-8B	60.0%	70.0%	+10.0 pp	2.2%	30.0%
GPT series / API					
GPT-5.6-Luna (Medium)	86.7%	100.0%	+13.3 pp	13.3%	45.6%
GPT-5.6-Luna (Instant)	81.1%	100.0%	+18.9 pp	10.0%	51.1%
GPT-5.4-Mini (Medium)	88.9%	100.0%	+11.1 pp	8.9%	38.9%
GPT-5.4-Mini (Instant)	64.4%	100.0%	+35.6 pp	11.1%	41.1%
GPT-5.4-Nano (Medium)	85.6%	97.8%	+12.2 pp	7.8%	36.7%
GPT-5.4-Nano (Instant)	62.2%	98.9%	+36.7 pp	4.4%	47.8%
Average	**72.5%**	**94.3%**	**+21.8 pp**	**5.6%**	**35.9%**

B.5 Token Costs

In Table 10, we report mean total tokens per completed response across each evaluation condition. Across the direct-inference benchmark, raw table and text inputs required an average of 11.4k and 13.2k tokens, compared with 2.4k and 2.9k for processed inputs, respectively. Averaged across both representations, precomputed summaries reduced total token usage by 78.6% compared with raw particle records. Serializing measurements as text further increased token usage relative to tabular inputs, by 15.0% for raw records and 23.6% for processed summaries. Analysis-routing tasks required 1.3k tokens per response on average, while the open-ended top-3 study required 16.7k, reflecting the larger set of morphology measurements provided to the model. These results show that shifting numerical computation away from the LLM through precomputed summaries or deterministic tools can also reduce the token requirements of LLM-guided characterization.

Table 10: **Mean total tokens by evaluation condition.** Values are mean total tokens per completed response, grouped with the same model order and task families as Table 1.

	Direct answer				Routing	Repeat test	Case study	
Model	Raw table	Raw text	Proc. table	Proc. text	Analysis routing	Repeat prompts	Pairwise comp.	Family top-3
Open-weight / local								
Gemma4 31B	14,326	13,598	3,463	3,384	1,312	8,199	827	18,077
Qwen3.6-35B-A3B	18,726	15,944	4,219	3,546	2,073	9,532	2,597	20,611
GPT-OSS-20B (Low)	11,004	13,496	1,841	2,754	1,353	5,958	1,534	17,341
GPT-OSS-20B (Medium)	11,205	13,240	1,853	2,759	1,382	5,720	2,087	19,281
GPT-OSS-20B (Default)	11,239	13,157	1,869	2,758	1,367	5,773	2,285	19,241
Gemma4-26B-A4B	14,346	13,599	3,473	3,385	1,307	8,201	827	18,105
Bonsai-8B	14,297	13,048	3,438	3,281	1,191	8,011	797	17,671
GPT series / API								
GPT-5.6-Luna (Medium)	9,070	12,629	1,731	2,648	1,132	5,224	578	14,491
GPT-5.6-Luna (Instant)	8,645	12,234	1,728	2,647	1,094	5,034	483	14,067
GPT-5.4-Mini (Medium)	9,236	12,777	1,758	2,677	1,192	5,349	721	14,956
GPT-5.4-Mini (Instant)	8,644	12,237	1,728	2,647	1,097	5,034	486	14,079
GPT-5.4-Nano (Medium)	9,238	12,761	1,745	2,653	1,201	5,322	753	14,757
GPT-5.4-Nano (Instant)	8,643	12,234	1,727	2,648	1,098	5,033	483	14,125

C Prompt Templates and Input Representations

C.1 Direct Inference Prompts

For direct inference, we use the same characterization prompt structure across four input representations, corresponding to the 2×2 factorial design described in Appendix A. Across representations, we include the eight morphological characteristics evaluated throughout this study, while processed inputs additionally contain the eight benchmarked summary statistics. We also include eight non-target morphological characteristics (`area_bbox`, `area_filled`, `equivalent_diameter_area`, `euler_number`, `perimeter`, `feret_diameter_max`, `solidity`, and `extent`). Processed inputs further include five non-target statistical measurements (mode, standard deviation, variance, coefficient of variation, and entropy), providing inputs where not all available information is directly relevant to the requested output. We provide representative examples below, with numerical values rounded for readability. Ellipses are used only in these examples to abbreviate long prompt sequences, while the prompts used during evaluation contained the complete serialized inputs without ellipses.

Processed-table prompt

```
Here is processed SEM powder morphology data in CSV table form.  What was the mean of the
area of the data?  Give only the number as an output with no other output.

Metric, Mean, Median, Mode, Range, Min, Max, IQR, Q1, Q3, ...
area, 9931.8, 5786.0, 3331.0, 16699, 3331.0, 20030, 10578, 4807.0, 15386, ...
area_bbox, 16973, 10050, 5487.0, 35722, 5487.0, 41209, 18962, 7880.0, 26842, ...
...
```

Processed-text prompt

```
Here is processed SEM powder morphology data embedded as text.  What was the mean of the
area of the data?  Give only the number as an output with no other output.

The Mean of area was 9931.8.  The Median of area was 5786.0.  The Mode of area was 3331.0.
The Range of area was 16699.  The Min of area was 3331.0.  The Max of area was 20030.
The IQR of area was 10578.  The Q1 of area was 4807.0.  The Q3 of area was 15386.  ...
The Mean of area_bbox was 16973.  The Median of area_bbox was 10050.  The Mode of
area_bbox was 5487.0.  The Range of area_bbox was 35722.  The Min of area_bbox was 5487.0.
The Max of area_bbox was 41209.  The IQR of area_bbox was 18962.  The Q1 of area_bbox was
7880.0.  The Q3 of area_bbox was 26842.  ...
...
```

Raw-table prompt

```
Here is per-particle SEM powder morphology data in CSV table form.  What was the mean of
the area of the data?  Give only the number as an output with no other output.

object_id, area, area_bbox, area_convex, area_filled, equivalent_diameter_area,
major_axis_length, minor_axis_length, eccentricity, euler_number, ...
0, 20030, 29524, 22280, 20030, 159.70, 245.06, 110.63, 0.89231, 1.0000, ...
1, 19764, 41209, 25361, 19764, 158.63, 191.07, 179.04, 0.34929, 1.0000, ...
...
```

Raw-text prompt

```
Here is per-particle SEM powder morphology data embedded as text.  What was the mean of
the area of the data?  Give only the number as an output with no other output.

Object 0 had an area of 20030, an area_bbox of 29524, an area_convex of 22280, an
area_filled of 20030, an equivalent_diameter_area of 159.70, a major_axis_length of
245.06, a minor_axis_length of 110.63, an eccentricity of 0.89231, and an euler_number of
1.0000.  ...
Object 1 had an area of 19764, an area_bbox of 41209, an area_convex of 25361, an
area_filled of 19764, an equivalent_diameter_area of 158.63, a major_axis_length of
191.07, a minor_axis_length of 179.04, an eccentricity of 0.34929, and an euler_number of
1.0000.  ...
...
```

C.2 Analysis-Routing and Comparative Characterization Prompts

For the remaining experiments, prompt structure varies based on the characterization task and the role assigned to the LLM. For analysis routing, prompts include the available tool schema, valid image identifiers, allowable morphology features and statistics, and a natural-language characterization objective. Models are instructed to select the appropriate tool and arguments without computing morphology values directly, returning a structured JSON call for deterministic execution.

For comparative characterization, models perform the specified analysis directly rather than selecting a tool. Pairwise prompts provide either raw per-particle measurements or precomputed characterization values for two images and ask the model to return the specified comparison. For top-3 comparisons, models are provided unordered deterministic summaries spanning 108 candidate morphology metrics and must identify three non-redundant morphology differences, along with the corresponding direction and interpretation. Unlike the top-k analysis-routing task, the model itself selects which morphology differences to report.

Analysis-routing prompt.

```
You are routing a natural-language SEM powder characterization request to one
deterministic tool. Do not calculate morphology values yourself.

{
  "task_type": "single_stat",
  "tool_schemas": [
    {
      "tool": "get_morphology_statistic",
      "arguments": {
        "image_id": "integer",
        "feature": ["equivalent_diameter", "circularity", "aspect_ratio", "elongation"],
        "statistic": ["percentile_10", "percentile_50", "percentile_90",
                      "median", "d90_minus_d10_over_d50"]
      }
    },
    ...
  ],
  "allowed_images": [
    {"image_id": 0, "image_name": "Cu-Ni-Powder_10_SE"},
    ...
  ],
  "routing_policies": [
    "Do not return clarify, none, or a prose refusal.",
    "The LLM selects the tool and arguments only; deterministic code computes values."
  ]
}

Return JSON only. Return exactly one object with keys tool and arguments.
Use only allowed IDs from the schemas and current task context.

User request: For Cu-Ni-Powder_10_SE, route a tool call for the typical
particle size.
```

Pairwise comparative prompt.

```
Here are per-particle SEM powder morphology values for two images as JSON.

{
  "A": {
    "image": "Cu-Ni-Powder_10_SE",
    "feature": "equivalent_diameter",
    "values": [
      159.70, 158.63, 143.39, 119.17, 79.605, 65.124, 136.45, 68.525, 80.685, ...
    ]
  },
  "B": {
    "image": "Cu-Ni-Powder_1_SE_V1",
    "feature": "equivalent_diameter",
    "values": [
      126.36, 38.974, 101.46, 86.636, 84.062, 132.16, 59.783, 52.685, 61.151, ...
    ]
  },
  "valid_answers": ["A", "B", "tie"]
}

Which image has the larger median equivalent diameter?
Answer only A, B, or tie.
```

Top-3 comparative prompt.

```
Here are deterministic SEM powder morphology summary facts for two images
as JSON. The metrics are intentionally unordered.

{
  "A": {
    "image": "S05_03_SE1_65",
    "metrics": [
      {"metric": "perimeter_cv", "value": 0.37880, "units": "unitless"},
      {"metric": "convexity_deficit_d90", "value": 0.10052, "units": "unitless"},
      {"metric": "equivalent_diameter_median", "value": 198.14, "units": "pixels"},
      {"metric": "area_d90", "value": 1.0608e5, "units": "pixels_squared"},
      {"metric": "solidity_cv", "value": 0.037360, "units": "unitless"},
      ...
    ]
  },
  "B": {
    "image": "Cu-Ni-Powder_10_SE",
    "metrics": [
      {"metric": "convexity_deficit_median", "value": 0.053731, "units": "unitless"},
      {"metric": "bounding_box_fill_median", "value": 0.71041, "units": "unitless"},
      {"metric": "aspect_ratio_iqr", "value": 0.41781, "units": "unitless"},
      {"metric": "equivalent_diameter_median", "value": 37.168, "units": "pixels"},
      {"metric": "solidity_cv", "value": 0.059728, "units": "unitless"},
      ...
    ]
  }
}

Valid metric IDs are: area_cv, area_d10, area_d90, area_iqr,
area_median, area_span_d90_d10_d50, area_bbox_cv,
area_bbox_d10, area_bbox_d90, ...

Compare image A and image B. Return JSON only: exactly 3 objects in an array.
Each object must have keys metric, larger, and interpretation. The metric must
be one of the valid metric IDs. larger must be A, B, or tie. Choose distinct
morphology categories with the largest differences; do not invent metrics.
```